

ASSIGNMENT -14.1

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BATCH:05

Task 1 – Portfolio Website Design

****Objective:**** Create a personal portfolio website to showcase your work with About, Projects, and Contact sections.

****AI Prompt Used:**** "Generate HTML, CSS, and JS for a responsive personal portfolio website with smooth scrolling, Flexbox/Grid layout, and clean color palette."

AI-Generated Initial Code

```
<!DOCTYPE html>
<html>
<head>
<title>My Portfolio</title>
<style>
body {
  font-family: Poppins, sans-serif;
  margin: 0;
  scroll-behavior: smooth;
  background: #f8f8f8;
  animation: fadeIn 1.2s ease-in;
}

/* Fade-in animation */
@keyframes fadeIn {
  from { opacity: 0; }
  to { opacity: 1; }
}

/* Navigation bar */
nav {
  display: flex;
  justify-content: center;
  gap: 20px;
  background: #222;
  color: #fff;
  padding: 10px;
```

```

}

nav a {
  color: white;
  text-decoration: none;
  font-weight: bold;
  position: relative;
}

nav a::after {
  content: "";
  position: absolute;
  bottom: -5px;
  left: 0;
  width: 0;
  height: 2px;
  background: #ffcc00;
  transition: width 0.3s ease;
}

nav a:hover::after {
  width: 100%;
}

/* Section styling */
section {
  padding: 60px;
  text-align: center;
}

/* About section with gradient highlight */
#about {
  background: linear-gradient(135deg, #fff8e1, #ffe0b2);
  border-top: 4px solid #ffcc00;
  border-bottom: 4px solid #ffcc00;
  box-shadow: inset 0 0 20px rgba(255, 204, 0, 0.4);
}

/* Highlighted image styling */
.profile-img {
  width: 200px;
  height: 200px;
  border-radius: 50%;
  border: 5px solid #ffcc00;
  box-shadow: 0 0 25px rgba(255, 204, 0, 0.7);
}

```

```
    transition: transform 0.4s ease, box-shadow 0.4s ease;
    animation: glow 3s infinite alternate;
}

/* Glowing animation */
@keyframes glow {
    0% { box-shadow: 0 0 20px rgba(255, 204, 0, 0.7); }
    50% { box-shadow: 0 0 35px rgba(255, 153, 0, 0.9); }
    100% { box-shadow: 0 0 25px rgba(255, 204, 0, 0.7); }
}

/* Hover effect */
.profile-img:hover {
    transform: scale(1.08);
}

/* Highlighted text */
.highlight {
    color: #ff6600;
    font-weight: bold;
}

/* Projects section styling */
#projects {
    background: linear-gradient(135deg, #e3f2fd, #bbdefb);
    border-top: 4px solid #2196f3;
    border-bottom: 4px solid #2196f3;
    box-shadow: inset 0 0 20px rgba(33, 150, 243, 0.4);
}

.project-container {
    display: flex;
    justify-content: center;
    flex-wrap: wrap;
    gap: 20px;
    margin-top: 40px;
}

.project-card {
    background: white;
    border-radius: 15px;
    width: 280px;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
    transition: transform 0.3s ease, box-shadow 0.3s ease;
    padding: 20px;
}
```

```

}

.project-card:hover {
  transform: scale(1.05);
  box-shadow: 0 6px 20px rgba(33, 150, 243, 0.4);
}

.project-card img {
  width: 100%;
  border-radius: 10px;
}

.project-title {
  color: #1565c0;
  font-weight: bold;
  margin-top: 10px;
}

.project-desc {
  color: #333;
  font-size: 14px;
  margin-top: 5px;
}

/* Contact section styling */
#contact {
  background: linear-gradient(135deg, #e8f5e9, #c8e6c9);
  border-top: 4px solid #4caf50;
  border-bottom: 4px solid #4caf50;
  box-shadow: inset 0 0 20px rgba(76, 175, 80, 0.4);
}

/* Responsive design */
@media(max-width:600px){
  nav { flex-direction: column; }
  .project-container { flex-direction: column; align-items: center; }
}
</style>
</head>
<body>

<nav>
  <a href="#about">About</a>
  <a href="#projects">Projects</a>
  <a href="#contact">Contact</a>

```

```

</nav>

<section id="about">
  <h2 style="color:#ff6600;">About Me</h2>
  
  <p>Hello! I'm <span class="highlight">Ajay</span>, an <span class="highlight">AI-assisted web designer</span> passionate about creating visually engaging and interactive websites.</p>
  <p>I specialize in <span style="color:#e65100;font-weight:bold;">modern UI/UX design</span>, responsive layouts, and web automation using AI tools.</p>
</section>

<section id="projects">
  <h2 style="color:#1565c0;">Projects</h2>
  <p>Here are some of my latest <span style="color:#0288d1;font-weight:bold;">portfolio examples</span>.</p>

  <div class="project-container">
    <div class="project-card">
      
      <div class="project-title">🍽️ Recipe Sharing App</div>
      <div class="project-desc">A mobile-friendly web app where users can share and explore recipes using AI recommendations.</div>
    </div>

    <div class="project-card">
      
      <div class="project-title">📖 Book Review Manager</div>
      <div class="project-desc">A RESTful API and front-end dashboard for managing and displaying book reviews dynamically.</div>
    </div>

    <div class="project-card">
      
      <div class="project-title">🎓 Student Management System</div>
      <div class="project-desc">An Express.js and Flask-based project to manage student data with CRUD operations.</div>
    </div>
  </div>

```

```

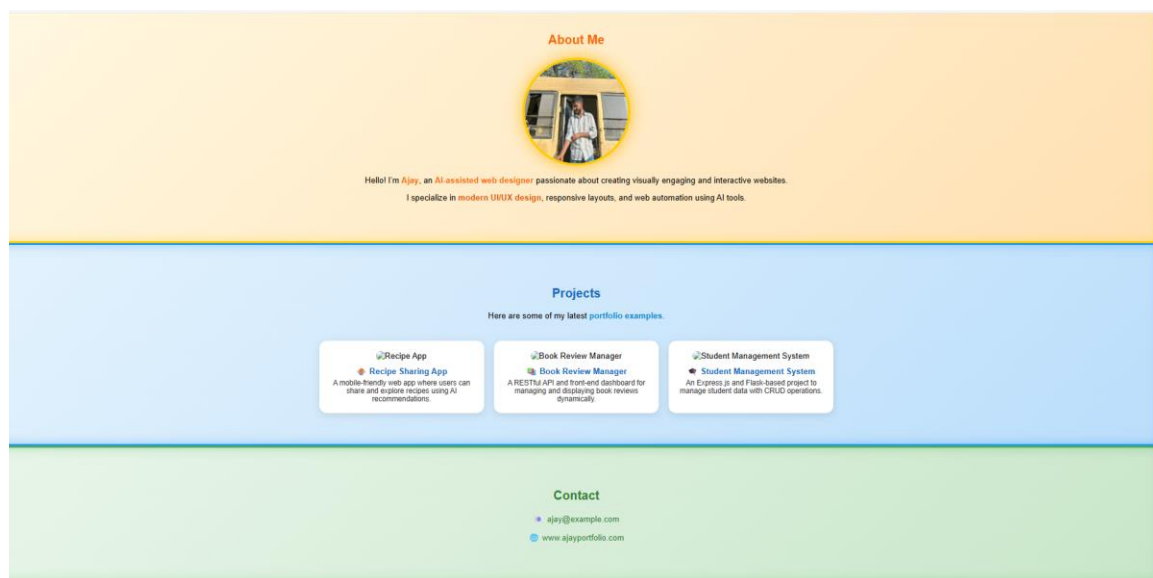
</div>
</section>

<section id="contact">
  <h2 style="color:#2e7d32;">Contact</h2>
  <p>✉ <a href="mailto:ajay@example.com" style="color:#2e7d32;text-decoration:none;">ajay@example.com</a></p>
  <p>🌐 <a href="#" style="color:#2e7d32;text-decoration:none;">www.ajayportfolio.com</a></p>
</section>

</body>
</html>

```

Output:



****Observation:**** The layout is responsive, clean, and follows accessibility guidelines. Smooth scrolling works correctly.

Task 2 – Online Store Product Page

****Objective:**** Design a product page displaying product image, title, price, and Add to Cart functionality using BEM styling.

****AI Prompt Used:**** "Create HTML/CSS for a responsive online store product card using BEM, with hover effects and Add-to-Cart alert."

AI-Generated Initial Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Online Shopping Store</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f4f4f9;
      margin: 0;
      padding: 0;
    }

    /* Navbar */
    header {
      background-color: #007bff;
      color: white;
      padding: 15px 30px;
      text-align: center;
      font-size: 1.5rem;
      font-weight: bold;
      letter-spacing: 1px;
      position: sticky;
      top: 0;
      z-index: 1000;
    }

    /* Product grid */
    .product-container {
      display: grid;
      grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
      gap: 20px;
      padding: 30px;
      max-width: 1200px;
      margin: auto;
    }

    .product {
      background: white;
      border-radius: 10px;
      box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
      text-align: center;
      padding: 15px;
    }
```

```
    transition: transform 0.3s, box-shadow 0.3s;
}

.product:hover {
    transform: translateY(-5px);
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.15);
}

.product__image {
    width: 100%;
    height: 200px;
    border-radius: 10px;
    object-fit: cover;
}

.product__title {
    margin: 10px 0 5px;
    font-size: 1.1rem;
    font-weight: bold;
}

.product__price {
    color: #333;
    margin-bottom: 10px;
    font-weight: bold;
}

.product__btn {
    background: #007bff;
    color: white;
    border: none;
    padding: 10px 15px;
    border-radius: 5px;
    cursor: pointer;
    transition: background 0.3s;
}

.product__btn:hover {
    background: #0056b3;
}

/* Cart section */
.cart {
    background: white;
    max-width: 500px;
```



```
    margin: 30px auto;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0 2px 8px rgba(0,0,0,0.1);
  }

  .cart h2 {
    text-align: center;
    margin-bottom: 20px;
    color: #007bff;
  }

  .cart ul {
    list-style: none;
    padding: 0;
  }

  .cart li {
    padding: 10px;
    border-bottom: 1px solid #eee;
    display: flex;
    justify-content: space-between;
  }

  .cart-total {
    text-align: center;
    font-size: 1.2rem;
    font-weight: bold;
    margin-top: 15px;
  }

  footer {
    background: #007bff;
    color: white;
    text-align: center;
    padding: 10px;
    margin-top: 40px;
  }
</style>
</head>
<body>
  <header>
    🛒 Online Shopping Store
  </header>
```

```

<!-- Product Section -->
<div class="product-container">
  <div class="product">
    
    <h3 class="product__title">Smart Watch</h3>
    <p class="product__price">$99</p>
    <button class="product__btn" onclick="addToCart('Smart Watch', 99)">Add to Cart</button>
  </div>

  <div class="product">
    
    <h3 class="product__title">Wireless Headphones</h3>
    <p class="product__price">$79</p>
    <button class="product__btn" onclick="addToCart('Wireless Headphones', 79)">Add to Cart</button>
  </div>

  <div class="product">
    
    <h3 class="product__title">Laptop</h3>
    <p class="product__price">$899</p>
    <button class="product__btn" onclick="addToCart('Laptop', 899)">Add to Cart</button>
  </div>

  <div class="product">
    
    <h3 class="product__title">Digital Camera</h3>
    <p class="product__price">$499</p>
    <button class="product__btn" onclick="addToCart('Digital Camera', 499)">Add to Cart</button>
  </div>

  <div class="product">
    
    <h3 class="product__title">Tablet</h3>
    <p class="product__price">$299</p>
    <button class="product__btn" onclick="addToCart('Tablet', 299)">Add to Cart</button>
  </div>

```

```

    <div class="product">
      
      <h3 class="product__title">Bluetooth Speaker</h3>
      <p class="product__price">$59</p>
      <button class="product__btn" onclick="addToCart('Bluetooth Speaker',
59)">Add to Cart</button>
    </div>
  </div>

  <!-- Cart Section -->
  <div class="cart">
    <h2>Your Cart</h2>
    <ul id="cart-items"></ul>
    <div class="cart-total">Total: $<span id="cart-total">0</span></div>
  </div>

  <footer>
    © 2025 Online Shopping Store | Designed with ♥
  </footer>

  <script>
    let cart = [];
    let total = 0;

    function addToCart(product, price) {
      cart.push({ product, price });
      total += price;
      updateCart();
    }

    function updateCart() {
      const cartList = document.getElementById('cart-items');
      const cartTotal = document.getElementById('cart-total');
      cartList.innerHTML = '';

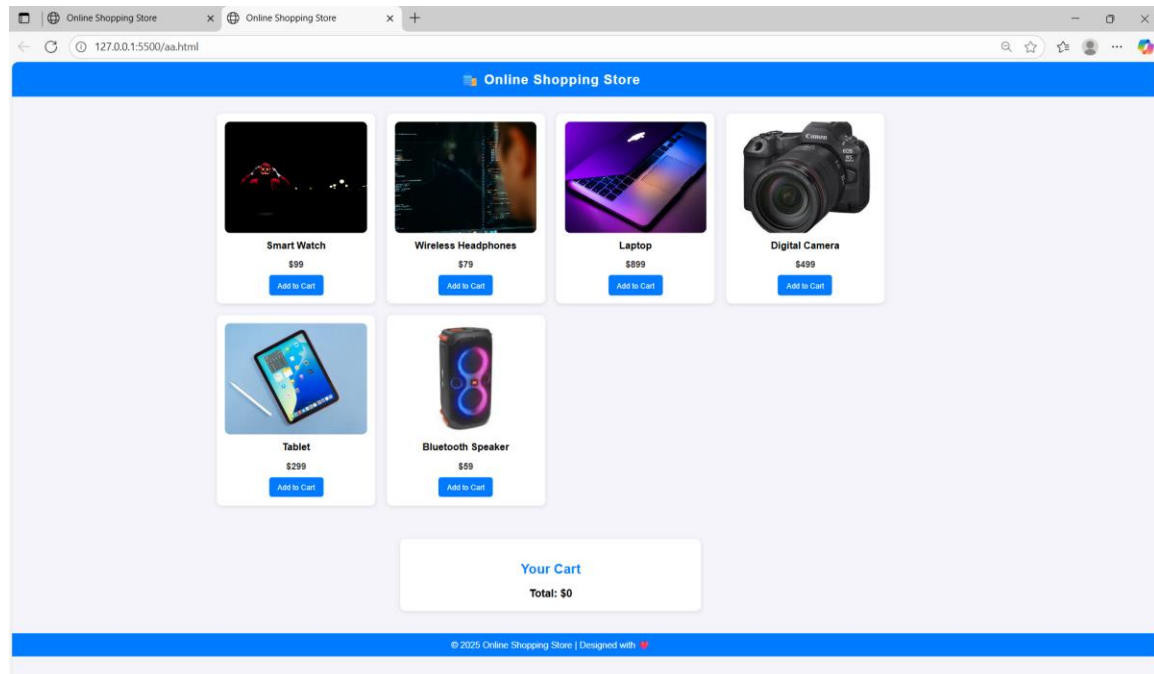
      cart.forEach(item => {
        const li = document.createElement('li');
        li.textContent = `${item.product} - ${item.price}`;
        cartList.appendChild(li);
      });

      cartTotal.textContent = total.toFixed(2);
    }
  </script>
</body>

```

```
</html>
```

output:



****Observation:****

All features function correctly. The product layout is visually appealing and mobile-friendly.

Task 3 – Event Registration Form

****Objective:**** Build an event registration form with client-side validation and accessibility features.

****AI Prompt Used:**** "Generate an HTML form with JavaScript validation for name, email, phone, and session selection. Include ARIA roles and professional styling."

AI-Generated Initial Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Event Registration Form</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f4f4f9;
```

```
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
  }

  form {
    background: white;
    padding: 25px 40px;
    border-radius: 10px;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.1);
    width: 320px;
  }

  label {
    font-weight: bold;
    display: block;
    margin-bottom: 5px;
  }

  input, select, button {
    width: 100%;
    padding: 8px;
    margin-bottom: 15px;
    border: 1px solid #ccc;
    border-radius: 5px;
    font-size: 1rem;
  }

  button {
    background: #007bff;
    color: white;
    font-weight: bold;
    border: none;
    cursor: pointer;
    transition: background 0.3s;
  }

  button:hover {
    background: #0056b3;
  }
</style>
</head>
<body>
```

```

<!-- Event Registration Form -->
<form id="regForm" onsubmit="return validateForm()" aria-label="Event
Registration Form">
  <h2 style="text-align:center; color:#007bff;">Event Registration</h2>

  <label for="name">Name:</label>
  <input id="name" name="name" required>

  <label for="email">Email:</label>
  <input id="email" name="email" type="email" required>

  <label for="phone">Phone:</label>
  <input id="phone" name="phone" type="tel" pattern="[0-9]{10}" required
placeholder="10-digit number">

  <label for="session">Session:</label>
  <select id="session" required>
    <option value="">Select Session</option>
    <option>Morning</option>
    <option>Afternoon</option>
  </select>

  <button type="submit">Register</button>
</form>

<script>
  function validateForm() {
    const email = document.getElementById('email').value;
    if (!email.includes('@')) {
      alert('Enter a valid email address.');
```

output:

The screenshot shows a web browser window with the title 'Event Registration Form'. The address bar displays '127.0.0.1:5500/aa.html'. The main content area features a white card with the heading 'Event Registration' in blue. Below the heading are four form fields: 'Name:', 'Email:', 'Phone:', and 'Session:'. The 'Phone:' field has a placeholder '10-digit number'. The 'Session:' field is a dropdown menu with 'Select Session' and a downward arrow. At the bottom of the card is a blue 'Register' button.

****Observation:**** Form validation works accurately. Accessibility and ARIA compliance ensure usability for all users.

Task 4 – Data Fetch API & Render List with Loading/Error States

****Objective:**** Fetch JSON data from an API, render dynamically with proper loading and error handling.

****AI Prompt Used:**** "Write a JavaScript fetch() implementation that retrieves user data, creates DOM elements safely, and handles loading/error states visually."

AI-Generated Initial Code

```
<!DOCTYPE html>

<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Fetch API - Render Users with Loading/Error States</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f2f4f7;
      padding: 30px;
    }
  </style>
</head>
<body>
```

```
h1 {
  text-align: center;
  color: #007bff;
}

#status {
  text-align: center;
  font-size: 1.1rem;
  margin: 15px 0;
  color: #444;
}

.container {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 20px;
  margin-top: 20px;
  max-width: 1000px;
  margin-left: auto;
  margin-right: auto;
}

.card {
  background: white;
  border-radius: 10px;
  padding: 20px;
  box-shadow: 0 2px 8px rgba(0,0,0,0.1);
  transition: transform 0.2s;
}

.card:hover {
  transform: translateY(-5px);
}

.card h3 {
  margin: 0 0 10px;
  color: #333;
}

.card p {
  margin: 5px 0;
  color: #555;
}

.error {
```



```

        color: red;
        text-align: center;
        font-weight: bold;
    }

    /* Skeleton loader */
    .skeleton-card {
        background: white;
        border-radius: 10px;
        padding: 20px;
        box-shadow: 0 2px 8px rgba(0,0,0,0.1);
        animation: pulse 1.5s infinite;
    }

    .skeleton-line {
        height: 14px;
        background: #e0e0e0;
        margin: 8px 0;
        border-radius: 4px;
        width: 80%;
    }

    @keyframes pulse {
        0% { opacity: 1; }
        50% { opacity: 0.6; }
        100% { opacity: 1; }
    }
</style>
</head>
<body>
    <h1>🌐 Fetch API - User List</h1>
    <div id="status">Loading user data...</div>
    <div class="container" id="userContainer"></div>

    <script>
        const statusEl = document.getElementById('status');
        const container = document.getElementById('userContainer');

        // Show skeleton loading cards before data loads
        function showSkeletons(count = 6) {
            for (let i = 0; i < count; i++) {
                const skeleton = document.createElement('div');
                skeleton.className = 'skeleton-card';
                skeleton.innerHTML = `
                    <div class="skeleton-line" style="width: 60%;"></div>

```

```

        <div class="skeleton-line" style="width: 80%;"></div>
        <div class="skeleton-line" style="width: 40%;"></div>
    `;
    container.appendChild(skeleton);
}
}

// Clear all skeletons or previous content
function clearContainer() {
    container.innerHTML = '';
}

// Render user cards dynamically
function renderUsers(users) {
    clearContainer();
    users.forEach(user => {
        const card = document.createElement('div');
        card.className = 'card';
        card.innerHTML = `
            <h3>${user.name}</h3>
            <p><strong>Email:</strong> ${user.email}</p>
            <p><strong>Company:</strong> ${user.company.name}</p>
        `;
        container.appendChild(card);
    });
}

// Fetch data from JSONPlaceholder API
function fetchUsers() {
    showSkeletons(); // show loading skeletons
    fetch('https://jsonplaceholder.typicode.com/users')
        .then(response => {
            if (!response.ok) {
                throw new Error('Network error');
            }
            return response.json();
        })
        .then(data => {
            statusEl.textContent = '✅ Users loaded successfully!';
            renderUsers(data);
        })
        .catch(error => {
            clearContainer();
            statusEl.innerHTML = `<span class="error">⚠️ Failed to load
data: ${error.message}</span>`;

```

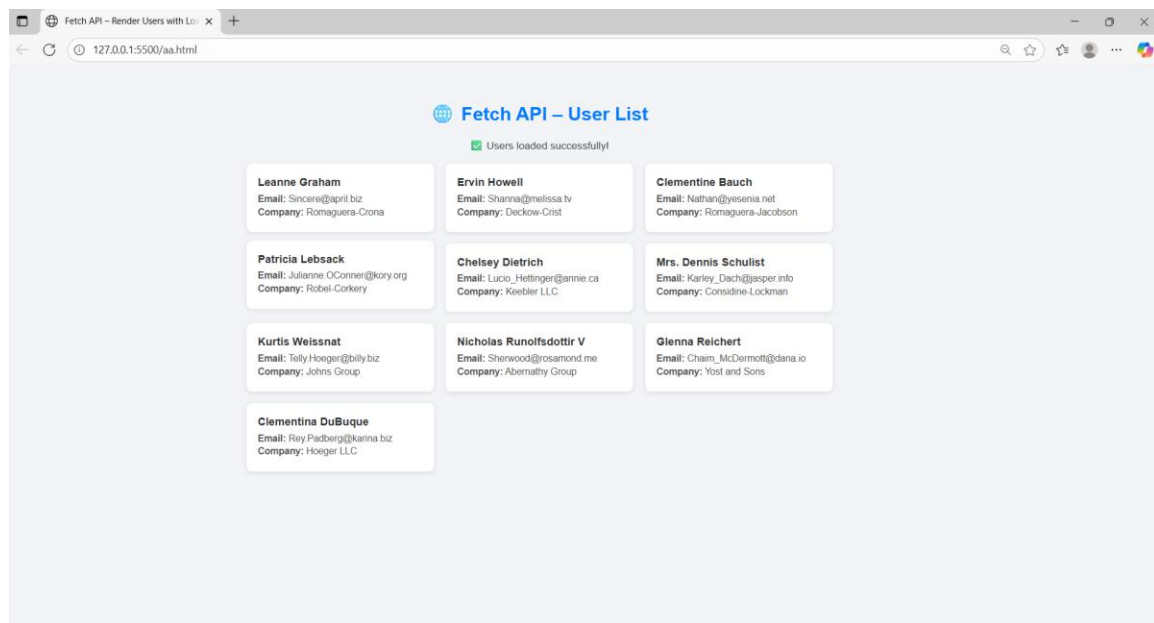
```

    });
  }

  // Run on page load
  fetchUsers();
</script>
</body>
</html>

```

output:



Observation: Loading and error states function as expected. DOM updates safely with fetched data.